

# IN PARTNERSHIP WITH PLYMOUTH UNIVERSITY

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## **PUSL3190 Computing Project**

# **Project Initiation Document**

## **SuwaMed: A Comprehensive Telehealth Platform**

*Bridging the Rural-Urban Healthcare Divide in Sri Lanka*

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HEAL SMART, FROM THE HEART

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## **Chapter 01 – Introduction**

### **1.1 Background and Context**

In Sri Lanka, the quality of health care varies considerably between urban and rural areas. Rural populations lack adequate health care infrastructure, qualified medical professionals, and reliable transportation to urban hospitals. This creates a serious health care gap where rural residents experience delayed diagnosis, inadequate treatment, and increased health risks.

Digital technology and mobile connectivity have created opportunities to democratize health care access. While mobile phone penetration extends even to rural areas of Sri Lanka, the healthcare sector has not taken full advantage of this infrastructure to address the urban-rural healthcare divide. Research indicates that properly designed telemedicine systems can provide rural populations equitable access to health care resources without the need for physical travel.

Current health care models rely heavily on physical presence, creating substantial barriers. Patients must take time off work, arrange transportation, and incur significant expenses for basic consultations. Meanwhile, qualified doctors remain concentrated in urban areas, with their expertise underutilized by rural populations. The COVID-19 pandemic further highlighted these disparities, demonstrating the urgent need for remote health care solutions.

### **1.2 Problem Statement**

When individuals experience clinical symptoms, they often lack immediate access to professional medical advice, leading to unreliable self-diagnosis or delayed consultation. The cost and time required to travel to urban health facilities pose considerable barriers, particularly for elderly patients, people with mobility problems, or families with limited resources.

While telemedicine has emerged globally, existing platforms fail to meet the specific needs of Sri Lankan rural communities. Many international telehealth applications require high-speed Internet, require digital literacy, and lack local language support. These platforms rarely integrate culturally appropriate interfaces or account for the unique health care ecosystem in Sri Lanka, including local payment systems, endemic diseases such as dengue, and doctor-patient communication styles.

### **1.3 Scope and Limitations**

#### **Features Included:**

- Dual-platform application (Flutter Mobile and React Web)
- AI symptom checker supporting Sinhala, Tamil, and English
- Secure video and text consultation optimized for low bandwidth (2G/3G)
- Appointment scheduling system for patients and doctors
- Encrypted electronic health records with document upload

- Multi-tier membership system with local payment integration (LankaQR, iPay)
- Doctor profiles with specialization, availability, and fee management
- Patient and doctor dashboards with history tracking
- Rating and review system for quality assurance
- Push notifications, in-app chat, and digital prescription generation

**Features Excluded:**

- Hospital Management System Integration
- Laboratory test ordering and medical imaging viewing
- Wearable device integration
- Insurance claim processing
- Offline consultation mode
- AI-powered diagnostics (only initial assessment provided)

## **1.4 Expected Impact and Stakeholders**

**Major Effects of the Current Problem:**

- Delayed diagnosis and treatment leading to poor health outcomes
- High costs associated with travel to urban health facilities
- Limited access to specialist advice for chronic conditions
- Lack of health information and reliance on unreliable self-diagnosis
- Underutilization of qualified doctors who could serve larger populations remotely

**Expected Outcomes:**

- Fully functional mobile apps (iOS/Android) and responsive web application
- AI symptom checker operational in three languages
- Secure infrastructure supporting 50+ concurrent consultations
- Encrypted medical records compliant with data security standards
- Sustainable subscription-based revenue model
- 25+ registered doctors and 250+ patients are targeted in the pilot phase
- Platform optimized for 2G/3G connections with 99.5% uptime during business hours

## Chapter 02 – Business Case

### 2.1 Business Need

The lack of a comprehensive, locally focused telehealth platform means that rural communities in Sri Lanka suffer from preventable health complications and limited access to timely medical intervention. The project addresses the critical health care access challenges faced by millions of Sri Lankans.

#### Key Business Drivers:

- Rural populations face 2-4 hour travel times for basic medical consultations
- Average consultation cost including travel expenses exceeds LKR 2,000 per visit
- Existing international platforms (oDoc, Practo, WebMD) do not address local language needs or low-bandwidth environments
- Over 65% of Sri Lanka's population resides outside major urban centers with limited healthcare access
- Post-pandemic awareness has increased demand for remote healthcare solutions

### 2.2 Business Objectives

1. Reduce average patient travel time for consultations from 2-4 hours to under 15 minutes through remote access
2. Achieve platform accessibility for users on 2G/3G networks while maintaining video quality at minimum 360p
3. Enable doctors to expand their practice beyond geographical boundaries, increasing patient reach by 300%
4. Establish a sustainable subscription-based revenue model with tiered pricing for different user segments
5. Integrate local payment gateways (LankaQR, iPay, card payments) achieving 95% transaction success rate
6. Build user trust through verified doctor credentials and patient rating system achieving minimum 4.0 average rating

### 2.3 Financial Overview

The estimated project development cost is LKR 175,000, with potential optimization up to LKR 75,000 through the use of cost-effective cloud providers. The subscription-based model ensures long-term sustainability while maintaining affordability for the rural population.

| Item Description             | Cost (LKR) |
|------------------------------|------------|
| Firebase Services (5 months) | 45,000     |

|                                       |                |
|---------------------------------------|----------------|
| Video Consultation API (Agora.io)     | 30,000         |
| Payment Gateway Integration           | 15,000         |
| UI/UX Design Tools (Figma - 5 months) | 8,000          |
| Cloud Hosting (AWS/Google Cloud)      | 25,000         |
| SSL Certificates and Domain           | 5,000          |
| Testing Devices (Android)             | 20,000         |
| AI/ML Training Resources              | 15,000         |
| Developer Tools and Plugins           | 7,000          |
| Documentation Tools                   | 5,000          |
| <b>Total Estimated Cost</b>           | <b>175,000</b> |



## Chapter 03 – Project Objectives

### 3.1 Functional Deliverables

#### **Objective 1: Develop a Cross-Platform Telehealth Application**

Build a mobile application for iOS and Android using Flutter, and a web application using React with a Firebase backend for real-time data synchronization, authentication, and cloud storage. The system will support responsive design across all device types.

#### **Objective 2: Implement AI-Powered Symptom Checker**

Develop an intelligent symptom assessment system with structured questionnaires in Sinhala, Tamil and English. The system will provide initial health assessments and suggest when professional consultation is needed, trained on Sri Lankan disease patterns including endemic conditions like dengue.

#### **Objective 3: Integrate Secure Video and Chat Consultation**

Implement end-to-end encrypted video conferencing and messaging using WebRTC/Agora.io, optimized for low-bandwidth connections (2G/3G). Enable real-time doctor-patient consultation with digital prescription capabilities.

#### **Objective 4: Create Comprehensive Appointment Management**

Develop a two-way platform that allows patients to find and book consultations based on expertise, language, ratings and fees, while enabling doctors to manage calendars, schedule availability and set consultation fees.

#### **Objective 5: Develop Secure Electronic Health Record Storage**

Implement AES-256 encrypted health record storage with role-based access controls. Patients can maintain health history and upload documents, while doctors can access relevant information during consultation.

#### **Objective 6: Create Subscription-Based Payment System**

Integrate local payment gateways (LankaQR, iPay, card payments) supporting different subscription tiers with automated billing. The system will include income dashboards for doctors with payment tracking.

### 3.2 Quality Criteria and Success Metrics

- Support for 50+ concurrent video consultations
- Response time less than 2 seconds for all operations
- Video quality maintained at 360p minimum on 2G/3G networks
- End-to-end encryption for all consultations
- AES-256 encryption for stored health records

- Role-based access control implementation
- 99.5% system uptime during business hours
- Android 7.0+ and iOS 12+ compatibility
- Web compatibility with all modern browsers
- Optimization for devices with minimum 2GB RAM

## **Chapter 04 – Literature Review**

### **4.1 Existing Solutions Analysis**

#### **oDoc (Sri Lanka)**

Sri Lanka's leading telehealth platform offers video consultations and home healthcare. However, it primarily targets urban populations with reliable internet access. It lacks an integrated AI symptom checker, has limited language support, and requires high bandwidth. Its cost structure may be prohibitive for rural populations.

#### **WebMD Symptom Checker**

Provides comprehensive symptom screening but is designed for Western healthcare systems. It lacks localization for Sri Lankan endemic diseases, has no local language support, and does not connect users to local doctors or integrate with local healthcare providers.

#### **Practo (India)**

Offers comprehensive healthcare services but faces integration challenges with Sri Lankan medical systems, payment gateways, and regulations. It is optimized for the Indian urban environment and does not address rural bandwidth challenges specific to Sri Lanka.

#### **Doctor on Call (Sri Lanka)**

Provides telephone consultation but lacks video capabilities, symptom assessment tools, and digital health record management, limiting its effectiveness for comprehensive health care delivery.

### **4.2 Identified Research Gaps**

#### **Knowledge Gap**

Limited research exists on integrated telehealth platforms combining AI symptom assessment, video consultation and health records specifically for Sri Lankan rural communities. Most research focuses on components independently rather than integrated, culturally appropriate systems.

#### **Technological Gap**

Current solutions do not adequately address Sri Lankan rural technological challenges: intermittent electricity, low bandwidth connections (2G/3G), and limited smartphone capabilities. There is a significant gap in developing applications that provide quality consultation over 2G while maintaining security and user experience.

#### **Methodological Gap**

Existing symptom checkers use rule-based algorithms from Western medical databases without machine learning adapted to local disease patterns and spatial conditions. Current platforms lack

mechanisms for continuous improvement through local medical data and do not account for Sri Lankan doctor-patient communication styles.

### **Contextual Gap**

International platforms are not designed with Sri Lankan regulatory requirements, payment systems or cultural expectations in mind. They do not integrate local payment gateways, lack Sinhala/Tamil support, and do not focus on the trust-building needed in communities where face-to-face health care has been the norm.

## **4.3 How SuwaMed Addresses These Gaps**

SuwaMed provides a local-centric ecosystem addressing the identified gaps. Unlike oDoc's urban approach, SuwaMed optimizes for low-bandwidth through adaptive video quality and 2G/3G compatibility. It offers full multilingual support with culturally appropriate medical terminology in Sinhala, Tamil, and English.

Unlike WebMD, SuwaMed's AI will train on Sri Lankan disease patterns, including endemic conditions like dengue. The Symptom Checker connects seamlessly to the doctor's consultation, creating an integrated health care journey rather than separate assessments.

Key innovations include: integrated ecosystem combining symptom screening, counseling and health records; Dual platform access (mobile and web); Affordable subscription model; low-bandwidth optimization; encrypted health records; and local payment integration. For doctors, it enables practice expansion beyond geographical boundaries with flexible scheduling, patient management, and earnings analysis.

## **Chapter 05 – Method of Approach**

### **5.1 Research Design**

The project follows a prototype development approach with iterative refinements based on user feedback. The development methodology combines agile principles with structured milestone delivery to ensure orderly progress while maintaining flexibility for improvement.

### **5.2 Technology Stack**

#### **Development Tools:**

- Flutter SDK for cross-platform mobile development (iOS/Android)
- React.js for web application frontend
- Firebase Console for backend services
- Android Studio and Xcode for mobile development
- Visual Studio Code as primary IDE
- UI/UX design with Figma
- version control with Git/GitHub
- API testing with Postman

#### **APIs and Services:**

- Firebase Authentication for user management
- Cloud Firestore for real-time database
- Firebase Cloud Storage for file storage
- WebRTC/Agora.io for video consultation
- PayHere/iPay for payment processing
- Firebase Cloud Messaging for notifications

#### **Programming Languages:**

- Dart for Flutter mobile development
- JavaScript/TypeScript for React web development
- Python for AI/ML symptom checker development
- SQL for database queries

### **5.3 Development Methodology**

The project employs an agile development methodology with two-week sprints, allowing iterative development and continuous improvement based on feedback from pilot users and stakeholders.

### **Sprint Structure:**

1. Requirements Refinement and Sprint Planning
2. development and implementation
3. Testing and quality assurance
4. Sprint review and retrospective

## **5.4 Special Knowledge Requirements**

- Cross-platform mobile development expertise
- Firebase architecture and real-time database design
- WebRTC protocols for video communication
- Payment gateway integration with local providers
- Machine learning for symptom assessment
- Healthcare Data Privacy and Security Standards
- UI/UX Design for Health Care Applications
- Low-bandwidth optimization techniques
- Medical terminology in Sinhala, Tamil and English

## **5.5 Data Collection and Validation**

Partnerships are being established with local healthcare providers and private practice doctors in Colombo and Gampaha districts for validation and pilot testing. Selected rural community centers in Kurunegala and Anuradhapura will serve as test sites for low-bandwidth performance evaluation.

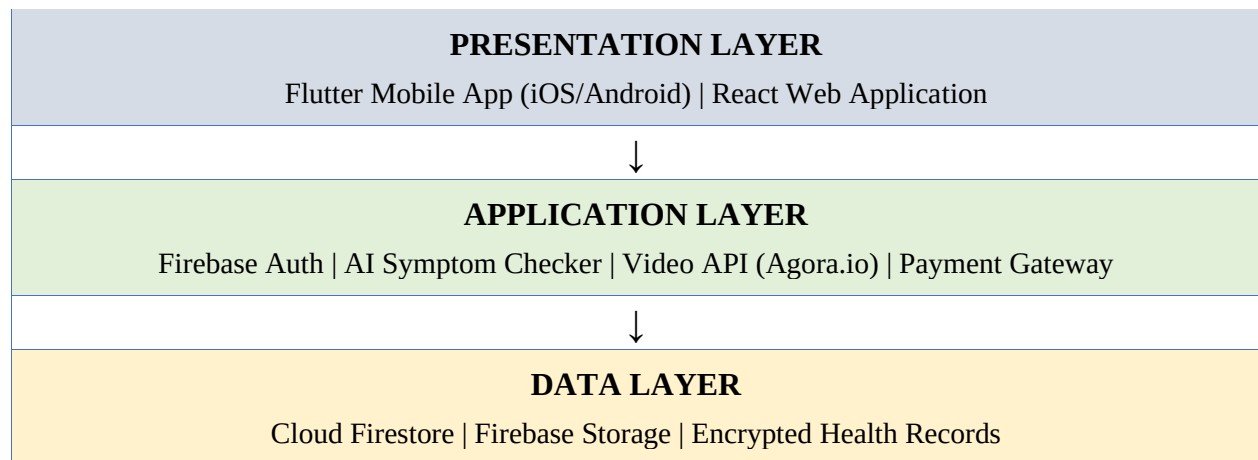
Technical integration support is being sought from payment gateway providers (PayHere and iPay). Medical advisory input will be obtained from qualified healthcare professionals to validate the symptom checker algorithms. User experience testing will include focus groups with urban and rural users representing different demographics, digital literacy levels, and language preferences.

## Chapter 06 – Conceptual Diagram

### 6.1 System Architecture Overview

The SuwaMed platform follows a three-tier architecture that includes the presentation layer (mobile and web applications), application layer (Firebase services and APIs), and data layer (Cloud Firestore and Firebase Storage).

#### High-Level Architecture Components:



### 6.2 Data Flow Description

#### Patient Flow:

7. User registers/logs in via mobile or web application
8. Completes AI symptom checker questionnaire in preferred language
9. Receives initial assessment and doctor recommendations
10. Searches and selects doctor based on expertise, rating, and availability
11. Books appointment and completes payment through local gateway
12. Conducts video/chat consultation with doctor
13. Receives digital prescription and follow-up notifications

#### Doctor Flow:

14. Registers with credential verification
15. Sets up profile, expertise areas, and consultation fees
16. Manages availability calendar

17. Receives appointment notifications
18. Accesses patient history and symptom checker results
19. Conducts consultation and writes digital prescriptions
20. Tracks earnings through income dashboard



## Chapter 07 – Initial Project Plan

### 7.1 Project Timeline

The SuwaMed project will be completed in five months (20 weeks) following a structured timeline to ensure systematic progress in research, design, development, testing, and documentation. The timeline allocates sufficient time to learn new technologies, implement complex features, conduct thorough testing, and prepare comprehensive documentation.

### 7.2 Project Phases and Milestones

| Phase                  | Duration    | Key Deliverables                                               |
|------------------------|-------------|----------------------------------------------------------------|
| Research & Planning    | Weeks 1-4   | Requirements analysis, literature review, system design        |
| Design                 | Weeks 3-6   | UI/UX design, database schema, API specification               |
| Development Setup      | Weeks 5-6   | Environment setup, Firebase configuration, project scaffolding |
| Core Development       | Weeks 6-12  | Authentication, symptom checker, user dashboards               |
| Communication Features | Weeks 10-14 | Video consultation, chat, notifications                        |
| Health Records         | Weeks 12-16 | EHR system, prescription generation, payment integration       |
| Testing                | Weeks 14-18 | Unit testing, integration testing, UAT, performance testing    |
| Documentation          | Weeks 18-20 | Technical documentation, user manuals, final report            |

### 7.3 Key Milestones

- Week 4: Requirements analysis completion
- Week 6: UI/UX design finalization
- Week 12: MVP (Minimum Viable Product) development
- Week 16: Testing phase completion
- Week 20: Final documentation and project submission

## Chapter 08 – Risk Analysis

### 8.1 Technical Risks

| Risk                                 | Likelihood | Impact | Mitigation Strategy                                                                     |
|--------------------------------------|------------|--------|-----------------------------------------------------------------------------------------|
| Video quality issues on 2G/3G        | High       | High   | Implement adaptive bitrate streaming; fallback to audio-only mode                       |
| Firebase service limitations         | Medium     | Medium | Design for scalability; implement caching; plan for alternative backends                |
| AI symptom checker accuracy          | Medium     | High   | Extensive testing with medical professionals; clear disclaimers; continuous improvement |
| Payment gateway integration failures | Medium     | High   | Multiple gateway support; offline payment verification; transaction retry logic         |
| Cross-platform compatibility issues  | Medium     | Medium | Comprehensive testing on multiple devices; responsive design principles                 |

### 8.2 Project Management Risks

| Risk                                | Likelihood | Impact | Mitigation Strategy                                                        |
|-------------------------------------|------------|--------|----------------------------------------------------------------------------|
| Schedule delays                     | Medium     | High   | Buffer time in schedule; prioritize MVP features; regular progress reviews |
| Learning curve for new technologies | High       | Medium | Early technology exploration; online courses; prototype development        |
| Scope creep                         | Medium     | Medium | Clear scope definition; change control process; stakeholder communication  |
| Pilot user recruitment              | Medium     | Medium | Early engagement with healthcare providers; incentive programs             |

### 8.3 External Risks

- Regulatory changes in healthcare data privacy: Monitor policy developments; design for compliance flexibility
- Payment gateway policy changes: Maintain relationships with multiple providers; implement abstraction layer
- Internet infrastructure limitations: Design for graceful degradation; implement offline-capable features where possible

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